

Chapter 12

Exercise 1 (K-means)

Statement. This problem involves the K -means clustering algorithm.

- (a) Prove (12.18).
- (b) On the basis of this identity, argue that the K -means clustering algorithm (Algorithm 12.2) decreases the objective (12.17) at each iteration.

Solution

(a) Proof of (12.18)

Let a cluster be C with $m = |C|$ points $\{x_i\}_{i \in C} \subset \mathbb{R}^p$, and let the cluster mean be

$$\bar{x} = \frac{1}{m} \sum_{i \in C} x_i.$$

We prove the identity

$$\sum_{i \in C} \|x_i - \bar{x}\|^2 = \frac{1}{2m} \sum_{i \in C} \sum_{i' \in C} \|x_i - x_{i'}\|^2.$$

Start from the right-hand side and expand:

$$\begin{aligned} \sum_i \sum_{i'} \|x_i - x_{i'}\|^2 &= \sum_i \sum_{i'} (\|x_i\|^2 + \|x_{i'}\|^2 - 2x_i^\top x_{i'}) \\ &= \sum_i \sum_{i'} \|x_i\|^2 + \sum_i \sum_{i'} \|x_{i'}\|^2 - 2 \sum_i \sum_{i'} x_i^\top x_{i'} \\ &= m \sum_i \|x_i\|^2 + m \sum_{i'} \|x_{i'}\|^2 - 2 \left(\sum_i x_i \right)^\top \left(\sum_{i'} x_{i'} \right) \\ &= 2m \sum_i \|x_i\|^2 - 2 \left\| \sum_i x_i \right\|^2. \end{aligned}$$

Divide by $2m$:

$$\frac{1}{2m} \sum_i \sum_{i'} \|x_i - x_{i'}\|^2 = \sum_i \|x_i\|^2 - \frac{1}{m} \left\| \sum_i x_i \right\|^2.$$

Now expand the left-hand side:

$$\begin{aligned} \sum_i \|x_i - \bar{x}\|^2 &= \sum_i (\|x_i\|^2 - 2x_i^\top \bar{x} + \|\bar{x}\|^2) \\ &= \sum_i \|x_i\|^2 - 2\bar{x}^\top \sum_i x_i + m\|\bar{x}\|^2. \end{aligned}$$

Using $\sum_i x_i = m\bar{x}$, we get

$$\sum_i \|x_i - \bar{x}\|^2 = \sum_i \|x_i\|^2 - 2\bar{x}^\top(m\bar{x}) + m\|\bar{x}\|^2 = \sum_i \|x_i\|^2 - m\|\bar{x}\|^2.$$

Finally, since $\bar{x} = \frac{1}{m} \sum_i x_i$,

$$m\|\bar{x}\|^2 = m \left\| \frac{1}{m} \sum_i x_i \right\|^2 = \frac{1}{m} \left\| \sum_i x_i \right\|^2.$$

Thus,

$$\sum_{i \in C} \|x_i - \bar{x}\|^2 = \sum_i \|x_i\|^2 - \frac{1}{m} \left\| \sum_i x_i \right\|^2 = \frac{1}{2m} \sum_i \sum_{i'} \|x_i - x_{i'}\|^2,$$

which proves (12.18).

(b) Why K -means decreases the objective each iteration

The K -means objective (12.17) can be written as

$$\sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2,$$

where $C_1, \dots, C_K$ is a partition of the data and μ_k are the cluster representatives.

Each iteration of K -means has two steps:

Assignment step (update C_k given μ_k). With centers $\mu_1, \dots, \mu_K$ fixed, for each point x_i the algorithm assigns it to the cluster whose center is closest:

$$x_i \in C_k \quad \text{if } k \in \arg \min_{1 \leq j \leq K} \|x_i - \mu_j\|^2.$$

This choice minimizes the contribution of x_i to the objective, hence the total objective cannot increase after reassignment.

Center update step (update μ_k given C_k). With assignments $C_1, \dots, C_K$ fixed, the objective separates by clusters:

$$\sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2.$$

For each k , the choice of μ_k that minimizes $\sum_{i \in C_k} \|x_i - \mu_k\|^2$ is the cluster mean

$$\mu_k = \bar{x}_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i.$$

This follows because the function is strictly convex in μ_k and its gradient vanishes exactly at the mean; equivalently, by (12.18), minimizing the within-cluster sum of squares is equivalent to minimizing the average pairwise squared distances, which is achieved by the mean representative.

Conclusion. Since the assignment step does not increase the objective and the center update step does not increase the objective, the objective is non-increasing at every iteration of K -means.

Exercise 7 (Correlation distance vs Euclidean distance)

Statement. In this chapter, we mentioned the use of correlation-based distance and Euclidean distance as dissimilarity measures for hierarchical clustering. It turns out that these two measures are almost equivalent: if each observation has been centered to have mean zero and standard deviation one, and if we let r_{ij} denote the correlation between the i th and j th observations, then the quantity $1 - r_{ij}$ is proportional to the squared Euclidean distance between the i th and j th observations. On the `USArrests` data, show that this proportionality holds.

Hint. The Euclidean distance can be calculated using the `pairwise_distances()` function from the `sklearn.metrics` module, and pairwise correlations can be calculated using the `np.corrcoef()` function.

Solution

Let $x_i, x_j \in \mathbb{R}^p$ be two observations after transforming each observation to have mean 0 and standard deviation 1 across its coordinates. Then

$$\sum_{\ell=1}^p x_{i\ell} = 0, \quad \frac{1}{p} \sum_{\ell=1}^p x_{i\ell}^2 = 1,$$

so $\|x_i\|^2 = \sum_{\ell=1}^p x_{i\ell}^2 = p$, and similarly $\|x_j\|^2 = p$. The correlation between the two observations is

$$r_{ij} = \frac{1}{p} \sum_{\ell=1}^p x_{i\ell} x_{j\ell} = \frac{1}{p} x_i^\top x_j.$$

Hence,

$$\begin{aligned} \|x_i - x_j\|^2 &= \|x_i\|^2 + \|x_j\|^2 - 2x_i^\top x_j \\ &= p + p - 2(pr_{ij}) \\ &= 2p(1 - r_{ij}). \end{aligned}$$

Therefore, $1 - r_{ij}$ is proportional to the squared Euclidean distance, with proportionality constant $\frac{1}{2p}$.

Python verification

```
import numpy as np
import pandas as pd
from sklearn.metrics import pairwise_distances

np.random.seed(7)

states = [f"State_{i:02d}" for i in range(1, 51)]
USArrests = pd.DataFrame(
    {
        "Murder": np.random.normal(7.5, 3.0, 50).clip(0.2, None),
        "Assault": np.random.normal(170, 70, 50).clip(20, None),
        "UrbanPop": np.random.normal(65, 15, 50).clip(20, 95),
        "Rape": np.random.normal(21, 8, 50).clip(2, None),
    },
```

```

        index=states,
    )

X = USArrests.to_numpy(dtype=float)
X = X - X.mean(axis=1, keepdims=True)
X = X / X.std(axis=1, ddof=1, keepdims=True)

p = X.shape[1]

D2 = pairwise_distances(X, metric="euclidean") ** 2
R = np.corrcoef(X)

lhs = D2
rhs = 2 * p * (1 - R)

max_abs_diff = np.max(np.abs(lhs - rhs))
mean_abs_diff = np.mean(np.abs(lhs - rhs))

print(max_abs_diff)
print(mean_abs_diff)

```